

PhD Qualification Exam in Applied Mathematics

Mock Examination No. 4

Total: 100 points

Problems 1–5 are required. Solve exactly one of Problems 6 and 7. If both are attempted, only the higher score will be counted.

- 1. (15 points)** Let S be the natural cubic spline interpolating data (x_i, f_i) on $a = x_0 < \cdots < x_n = b$. Prove that for every $g \in H^2(a, b)$ satisfying $g(x_i) = f_i$,

$$\int_a^b |S''(x)|^2 dx \leq \int_a^b |g''(x)|^2 dx.$$

Prove the equality case. Is it true that a natural cubic spline reproduces every quadratic polynomial? Give a rigorous answer.

- 2. (15 points)** Consider $x_{k+1} = Mx_k + c$.

- Prove that the iteration converges to the same fixed point for every starting vector if and only if $\rho(M) < 1$.
- Give a 2×2 example with $\rho(M) < 1$ but $\|M\|_2 > 1$, and explain why this does not contradict convergence.
- Let G be continuously differentiable near a fixed point x_* . Show that $\rho(G'(x_*)) < 1$ implies local convergence, even though G need not be a contraction in the Euclidean norm.

- 3. (20 points)** Let

$$A_\alpha = \begin{bmatrix} 1 & \alpha \\ 0 & 1 \end{bmatrix}, \quad b = e_2, \quad x_0 = 0.$$

- Show that all eigenvalues equal 1 and that GMRES nevertheless terminates in at most two steps.
- Compute the first GMRES iterate and prove $\|r_1\|_2 / \|r_0\|_2 = |\alpha| / \sqrt{1 + \alpha^2}$.
- Explain why eigenvalue clustering alone does not predict GMRES convergence for nonnormal matrices.
- State the Arnoldi relation and the small least-squares problem used by GMRES, including the principal cost per iteration.

- 4. (15 points)** Consider a conservative three-point explicit scheme for $u_t + au_x = 0$, $a > 0$,

$$U_j^{n+1} = \alpha U_{j-1}^n + \beta U_j^n + \gamma U_{j+1}^n, \quad \nu = \frac{a\tau}{h} \in (0, 1).$$

Derive the moment conditions for consistency and second-order accuracy. Prove that no such second-order scheme can be monotone. Identify the second-order scheme and compare it with the monotone first-order upwind scheme.

- 5. (15 points)** For $-(a(x)u')' + c(x)u = f$ on $(0, 1)$ with $u(0) = u(1) = 0$, assume $a(x) \geq a_0 > 0$ and $c(x) \geq 0$. Formulate the conforming piecewise-linear finite element method. Prove Galerkin orthogonality and the energy-norm best-approximation property. If $u \in H^2(0, 1)$ and the mesh is shape regular, derive the $O(h)$ energy-norm error estimate.

- 6. (20 points; choose one of Problems 6–7)** Let $\phi \in C^2[-1, 1]$ and

$$I_a(\varepsilon) = \int_{-1}^1 \exp\left(-\frac{(x^2 - a^2)^2}{\varepsilon}\right) \phi(x) dx.$$

For fixed $0 < a < 1$, derive the leading contribution from all dominant points. Then treat the degenerate case $a = 0$, including the exact Gamma-function constant. Explain why the fixed- a expansion is not uniform as $a \downarrow 0$, and identify the transition scale.

- 7. (20 points; choose one of Problems 6–7)** Let $Q = Q^T > 0$, $c \in \mathbb{R}^n$, and $R > 0$. Consider

$$\min_{\|x\|_2 \leq R} \frac{1}{2} x^T Q x - c^T x.$$

Derive necessary and sufficient KKT conditions. If the unconstrained minimizer lies outside the ball, prove that there is a unique $\lambda > 0$ such that

$$x(\lambda) = (Q + \lambda I)^{-1} c, \quad \|x(\lambda)\|_2 = R.$$

Give a reliable numerical algorithm and its cost after an eigendecomposition of Q .